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# The neural correlates of noun and verb processing

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## Abstract

Aim of this paper is to review the neuropsychological and neuroimaging studies dealing with the neural correlates of noun and verb processing. Available evidence indicates that, while the left temporal neocortex plays a crucial role in all tasks involving lexical-semantic processing, additional regions of the left dorsolateral prefrontal cortex are recruited during the processing of words related to actions. This anatomo-clinical correlation, related to object and action naming, is found in many instances of ‘verb–noun’ dissociation reported in the neuropsychological literature. More recent investigations, which use a wider range of verb categories, and attempt to take into account other dimensions related to the verb–noun distinction, are starting to contribute to a better understanding of the mechanisms underlying the observed neurological dissociation.

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## 1. Introduction

Are there different neural correlates for noun and verb processing in the human brain? If any, do these differences reflect an underlying semantic distinction (for example, between lexical items referring to objects and items referring to actions) or can they be attributed to syntactic or morphological differences between nouns and verbs?

The answer to the former question appears to be affirmative, on the basis of considerable evidence coming from different sources: it is this evidence that will be the focus of the present review paper. On the other hand, there is still no a clear-cut response to the latter question. Many of the claims about noun/verb dissociations are based on clinical and imaging studies which have actually contrasted the processing of names of objects and

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actions. Thus, these findings do not give unequivocal support to the contention that brain anatomy reflects grammatical distinctions. This is a crucial point, considering that, while there is independent evidence for the presence of differences in the pattern of brain activation related to multiple semantic categories (for example, living vs. non living, Damasio, Grabowski, Tranel, Hichwa, & Damasio, 1996; Martin, Wiggs, Ungerleider, & Haxby, 1996; Perani et al., 1995), distinctions in brain activity related to morphosyntactic factors have not been clearly identified at the single-word processing level.

The available evidence, coming from neuropsychological studies as well as from functional neuroimaging, will be summarised in this review, with the principal aim to provide suggestions for further research in this area.

## 2. Neuropsychological evidence

Hints to possible different localisation of the lesions associated with defective processing of nouns and verbs could be derived from group investigations in aphasic patients, which assessed the ability to name objects and actions. In a landmark investigation, Goodglass, Klein, Carey, and Jones (1966) showed that Broca's aphasics are more impaired in naming actions, fluent aphasics in naming objects. From this result, within the limits of the localisation of aphasic syndromes, it could be concluded that posterior (retrolandic) lesions affect object naming more than action naming, while an involvement of pre-rolandic areas appears to be required for the presence of defective action naming.

The limitations of this approach are evident. Besides the uncertainty of the anatomical localisation of the lesions as inferred from the clinical diagnosis of an aphasic syndrome, it must be underlined that the small number of items for each category of stimuli prevented any generalisation about the noun–verb effect. The latter limitation was overcome by subsequent reports (Miceli, Silveri, Nocentini, & Caramazza, 1988), which established, using adequate testing material, the existence of patients with selective, or relatively selective, disorders in naming and comprehension of nouns and verbs. These investigations did not address in particular the question of the anatomical correlates; however, an analysis of the lesion site in the individual case appears to support, with a few exceptions, the anatomical localisation suggested by the Goodglass et al. (1966) study. Most of the patients with selective disorders of noun processing had posterior lesions, in particular involving the left temporal lobe; verb impairment, on the other hand, was associated with damage extending or limited to the left prefrontal cortex. This correlation was explicitly proposed in an influential paper about the neural correlates of noun and verb retrieval by Damasio and Tranel (1993). In this report the lesion location was carefully assessed with magnetic resonance imaging (MRI). Patient Boswell, who was impaired only for nouns, had bilateral mesial and lateral temporal lesions; patient AN, showing a similar dissociation, had a left anterior temporal lesion. On the other hand, patient KJ had a left premotor frontal lesion and was selectively impaired in action naming. It must be acknowledged, however, that there is a number of reported cases, which represent 'exceptions' to this correlation. The lesion of patient Mario, reported by De Renzi and Di Pellegrino (1995), who had a sparing of action naming and verb generation, involved the

temporal lobe, but extended to the frontal cortex. Several patients with a disproportionate deficit in verb processing had lesions centred in the left parietal lobe (see, for example, Silveri & Di Betta, 1997).

A cautious conclusion which can be drawn from the focal lesions studies is that lesions circumscribed to the left anterior–inferior temporal lobe result in a disorder of object naming associated with sparing of action naming. On the other hand, patients with a selective difficulty in action naming and, maybe, also of other aspects of language processing involving the grammatical category of verbs are as a rule affected by lesions which involve the frontal and parietal part of the left perisylvian language areas. The temporal lobe, and in particular its anterior part, are usually spared in these patients. It is noteworthy that, within the category of nouns, an interesting pattern of anatomical correlation has been derived from the investigation of patients with category-specific semantic disorders. Patients with defective naming of animals and persons have usually lesions affecting the temporal lobe, in particular its anterior part, while tool naming impairments are associated with perisylvian damage in the temporo-parietal and frontal areas (Damasio et al., 1996; Saffran & Schwartz, 1994). Taken together, these findings point to a possible relationship between tool naming and action naming which is compatible with the recent neuroimaging evidence which will be discussed later.

Additional data can be derived from the investigation of conditions associated with progressive neurological dysfunction. Daniele, Giustolisi, Silveri, Colosimo, and Gainotti (1994) were the first to report that noun–verb dissociations could be observed also in patients with degenerative conditions. They reported two patients with frontal lobe atrophy who were more impaired in naming actions than in naming objects; one patient with left temporal atrophy showed the reverse dissociation. The pattern appears to be more complex in patients with less selective degenerative pathology. Cappa et al. (1998) reported that, while action naming was more impaired than object naming in patients with probable Alzheimer's disease, the discrepancy was much larger in patients with a clinical diagnosis of fronto-temporal dementia. These two conditions, which are associated with a distinct neuropathology at the microscopic level, usually differ quantitatively in the extent of involvement of the frontal lobe at the macroscopic level. These results are thus in agreement with the evidence from patients with localised lesions, suggesting a link between frontal involvement and action naming impairment.

### 3. Functional imaging

The investigation of the neural correlates of noun and verb processing in the normal brain using functional neuroimaging has provided a number of important observations, although there are considerable inconsistencies. Neurophysiological investigations have in general confirmed the presence of 'anterior' activity related to the processing of action verbs (Pulvermueller, Lutzenberger, & Preissl, 1999); however, the anatomical sensitivity of these methods is limited. More precise information can be expected from functional imaging studies using PET and functional magnetic resonance. As in the case of clinical studies, also in this area many investigations have been concerned, rather than with verbs, with the retrieval of knowledge about actions. Because of the early, influential PET study

by Petersen, Fox, Posner, Mintun, and Raichle (1989), which used as ‘semantic task’ the generation a verb associated with a presented noun, many studies of word retrieval have actually examined the neural correlates of the retrieval of action knowledge. These studies have reliably shown extensive dorsolateral frontal activations. However, grossly similar patterns of activation have also been observed with noun generation tasks (phonemic and semantic fluency (Paulesu et al., 1997)). One investigation which attempted the comparison between noun generation and verb generation revealed minor differences, involving the temporo-parietal and frontal areas (Warburton et al., 1996). An interesting comparison between colour naming and action naming was reported by Martin, Haxby, Lalonde, Wiggs, and Ungerleider (1995): the selective activations associated with the latter task involved the left fronto-parietal cortex, the middle temporal gyrus and the cerebellum.

Direct evidence for a noun–verb difference, extending beyond the retrieval of action knowledge which is typically elicited by the verb generation task, comes from a PET experiment, using lexical decision (Perani et al., 1999). The stimuli were, respectively, nouns referring to tools and psychological states, and manipulation and psychological verbs. The results indicated the existence of incompletely overlapping neurological substrates for verb and noun processing. However, it must be underlined that there was no double dissociation between frontal and temporal cortex, but only the presence of ‘verb specific’ areas (Broca’s, left middle temporal gyrus). Noun and verb processing equally activated the other areas, associated with the lexical task. No significant interactions between grammatical class and semantic content were observed, suggesting that the observed difference is verb-specific. Another PET experiment, however, failed to confirm the latter findings (Tyler, Russell, Fadili, & Moss, 2001): no differences were found between closely matched nouns and verbs, both in a lexical decision and in a semantic judgement task. How to explain this discrepancy? Tyler et al. (2001) suggest that inadequate controls for parameters such as word length, frequency, familiarity and imageability may account for some of the differences. This criticism does not apply to the Perani et al. experiment, in which the stimuli were carefully matched for the above parameters. It is noteworthy that we were able to achieve a comparable behavioural performance with nouns and verbs in terms of error rates and reaction times. Further studies are required to understand the contribution of technical factors, task-related differences and, maybe, also of language-specific features to the lack of consistency in the results.

#### 4. Conclusions

The findings summarised above lead to the conclusion that, while there is considerable evidence for the existence of a difference in the cerebral correlates of the processing of object nouns and action verbs, the possibility to generalise these results to a ‘grammatical class’ difference is still unclear.

The proposal that the difference between nouns and verbs cannot be reduced to a single factor, but is actually based on a continuum of differences at the semantic, syntactic and phonological level (Black & Chiat, 2002) has important implications also for the interpretation of the neurological findings.

Table 1  
Talairach coordinates of Broca's area activation in two different PET experiments

|                                                            |            |
|------------------------------------------------------------|------------|
| Lexical decision on verbs (Perani et al., 1999)            |            |
| Ba 46/9                                                    | – 28 28 28 |
| Ba 45/46                                                   | – 36 30 20 |
| Detection of morphosyntactic anomalies (Moro et al., 2001) |            |
| Ba 45                                                      | – 28 34 8  |

As for the role of semantic factors, the hypothesis that a strong determinant of the observed neurological differences is actually due to the action content of the verbs used in most clinical and imaging experiment is supported by the results of several investigations related to action observation and action representation. Recognition of actions is an ability which is highly developed in humans and non-human primates. In primate, mirror neurones have been described in the left rostral part of the inferior area 6 (the so called F5), which becomes active when the same action is actively performed by the monkey or when it is made by the experimenter and observed by the monkey (Gallese, Fadiga, Fogassi, & Rizzolatti, 1996). A comparable activation was found in humans while observing motor actions (grasping of meaningful objects with the right hand (Rizzolatti et al., 1996)). The same area was also involved within a network of brain structures, which became active during observation of meaningful pantomimes in comparison with meaningless gestures (Decety et al., 1997). More specifically, the left inferior frontal gyrus, namely Ba 44 and 45 (Broca's area), have been related to the 'representation' of hand actions. For example, Parsons et al. (1995) found functional activation of this region during a task in which normal subjects were required to form a mental image of the hand and rotate it. Decety et al. (1994), using PET, found activation of the inferior frontal gyrus during mental simulation of actions. The observation/execution matching system ('mirror neurones') identified both in monkeys and in man, can thus be considered as a putative system specialised both for the encoding and the production of actions, and may represent the basis of the recognition of meaningful motor events. Broca's area might be endowed both with a hand movement representation and action recognition system. The relationship between frontal areas and verb processing, according to this hypothesis, would thus be reduced to the overt engagement of motor representations in tasks such as action naming, and in its automatic recruitment during linguistic task which, in many of the investigations reviewed here, involved verbs related to hand movement.

There is, however, ground to suggest that other factors, besides semantic content, may be responsible for the noun–verb difference. Morphological factors should also be taken into account: for example, the presence of verb-specific suffixes in languages as Italian may be hypothesised to play a role in the experiment of Perani et al. (1999). It is noteworthy that the verb-specific area observed in the latter investigation is localised in a part of Broca's area which has been found to be activated by the detection of syntactic anomalies (Table 1), suggesting that the observed difference in activation may be related to the grammatical specificity of verbs. Among other dimensions, the number of arguments appears to play a role in determining the clinically observed dissociations. In three agrammatic patients, the noun–verb difference found with non-argumental nouns, such as medal, was not confirmed when argumental nouns, such as theft, were compared to

verbs (Collina, Marangolo, & Tabossi, 2001). These results indicate that the noun advantage observed in agrammatic patients is abolished if nouns and verbs are matched for argument structure. Future imaging investigations may help to clarify the respective contribution of the multiple different dimensions of verbs and nouns to the observed neurological differences in localisation.

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